INSTRUCTOR MANUAL

CSE 4571 — Practical Robotics Project with Arduino

Seventh Semester B.Tech (2L – 4P)| Hardware-Integrated / Wokwi Simulation-Based Laboratory

Practical Arduino Robotics — Lukas Kaul

SESSION-4

Stepper Motor Position and Direction Control using the A4988 Driver

Learning Domain Distribution — Cognitive: 30% | Psychomotor: 55% | Affective: 15%

Note: This experiment is conducted in the Wokwi browser-based simulation environment and, where hardware is available, on a physical Arduino UNO / A4988 / bipolar stepper motor bench setup. Simulation is used first to verify logic safely before any physical current is applied to the motor coils.

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1. Session Information

Field	Details
Course Code / Title	CSE 4571 — Practical Robotics Project with Arduino
Semester	Final Year B.Tech (CSE)
Course Format	Hardware-Integrated / Wokwi Online Simulation-Based Laboratory (STEP/DIR interfacing)
Session	Experiment 4
Duration	6 Hours (360 minutes)
Software and Hardware Used	Wokwi (Online Circuit Simulation Platform), Arduino IDE, A4988 driver module, bipolar stepper motor, and physical components
Topic	Stepper Motor Position and Direction Control using the A4988 Driver
Textbook Reference	Allegro A4988 DMOS Microstepping Driver Datasheet; Wokwi Simulation Documentation

Hardware components, apparatus, driver modules, and bipolar stepper motors are integrated where required, while simulation-based activities are performed on-screen using Wokwi.

2. Learning Objectives

By the end of this session, students will be able to:

Objectives	Statements
O1	Explain the working principle of a bipolar stepper motor and the function of a dedicated stepper driver IC.
O2	Describe the role of the STEP, DIR, ENABLE (and MS1/MS2/MS3, RESET, SLEEP) pins of the A4988 driver.
O3	Correctly interface an Arduino UNO, an A4988 driver, and a bipolar stepper motor, ensuring proper signal, power, and common-ground connections, in Wokwi and/or on hardware.
O4	Write and upload an Arduino sketch that generates STEP pulses and toggles DIR to reverse rotation.
O5	Relate a commanded step count to the resulting angular displacement of the motor shaft and verify it experimentally.

3. Course Outcomes (CO1–CO6)

CO	Course Outcome Statement	Learning Domain	Taxonomy Level
CO1	Explain the fundamentals of Arduino-based robotic systems and interface different sensors for	Cognitive (L) + Psychomotor (P) + Affective (A)	L2 – Understand+ P1: Perception+ A1: Receiving

	measurement, analog data acquisition, and real-time sensing applications.		
CO2	Develop Arduino programs to control robotic actuators including servo motors, stepper motors, and DC motors for position and motion-based applications.	Cognitive (L) + Psychomotor (P) + Affective (A)	L3 – Apply + P3 – Guided Response+ A2: Responding
CO3	Implement intelligent robotic control systems using sensor inputs, finite state machines, non-blocking programming techniques, and feedback-based control mechanisms.	Cognitive (L) + Psychomotor (P) + Affective (A)	L3 – Apply + P4 – Mechanism + A3: Valuing
CO4	Design modular and reusable robotic software components using object-oriented programming concepts and Arduino libraries for embedded applications.	Cognitive (L) + Psychomotor (P) + Affective (A)	L5 – Evaluate + P5 – Complex Overt Response+ A4: Organization
CO5	Analyze and troubleshoot robotic systems by integrating communication interfaces, remote-control techniques, and wireless monitoring using embedded platforms.	Cognitive (L) + Psychomotor (P) + Affective (A)	L4 – Analyze + P6 – Adaptation + A4: Organization
CO6	Design and demonstrate an Arduino-based robotic application by integrating sensors, actuators, control algorithms, and wireless communication modules while following engineering practices.	Cognitive (L) + Psychomotor (P) + Affective (A)	L6 – Create + P7 – Origination + A5 – Characterization

L: Cognitive (L1: Remembering; L2: Understanding; L3: Applying; L4: Analyzing; L5: Evaluating; L6: Creating)

P: Psychomotor (P1: Perception; P2: Set; P3: Guided Response; P4: Mechanism; P5: Complex Overt Response; P6: Adaptation; P7: Origination)

A: Affective (A1: Receiving; A2: Responding; A3: Valuing; A4: Organization; A5: Characterization)

UN Sustainable Development Goals (SDGs)

The course CSE 4571 - Practical Robotics Project with Arduino primarily reflects the following Sustainable Development Goals (SDGs):

SDG	Goal	Course Relevance
SDG 4	**Quality Education**	Develops practical skills in Arduino programming, stepper driver interfacing, STEP/DIR signal generation, position and direction control, circuit simulation, observation, data interpretation, and troubleshooting.
SDG 7	**Affordable and Clean Energy**	Promotes efficient operation of stepper-motor-based control systems by using correct driver-enable practices, current-limiting, and open-loop control, supporting awareness of energy-efficient embedded and automation systems.
SDG 9	**Industry, Innovation and Infrastructure**	Promotes innovation in robotic and automation systems through Arduino-based stepper motor position and direction control, with applications in 3D printers, CNC machines, robotic arms, and camera gimbals.

SDG 11	**Responsible Consumption and Production**	Supports smart and automated systems through precise open-loop position control, with applications in automated manufacturing, robotic assembly, and precision motion systems that reduce material waste.

CO – PO Mapping (Focus: CO1-CO6)

CO	PO Mapped	Strength
CO1	PO1, PO2	High
CO2	PO1, PO5	High
CO3	PO1, PO4, PO5	High
CO4	PO2, PO4, PO5	High
CO5	PO3, PO5, PO6	High
CO6	PO3, PO5, PO9, PO10, PO11	High

Day Focus for this session: CO2 → PO1, PO3, PO5 (High); CO3 → PO3, PO5, PO9 (High).

4. Learning Domain Distribution (3H Model)

The 360-minute session strictly maintains the prescribed session-level domain distribution of Cognitive 30%, Psychomotor 55%, and Affective 15%.

Domain	Weightage	Time (approx.)	Focus
Cognitive (Head)	30%	108 minutes	Bipolar stepper motor theory, the role of a dedicated driver IC, A4988 STEP/DIR/ENABLE/MS pin functions, and the steps-to-angle relationship.
Psychomotor (Hand)	55%	198 minutes	Wiring the A4988 driver and stepper motor to the Arduino UNO; writing and uploading the STEP/DIR control sketch; running and observing the simulation; modifying step counts; troubleshooting wiring and code errors.
Affective (Heart)	15%	54 minutes	Safe-wiring discipline, peer verification of connections, systematic observation of motor response, and lab etiquette during hardware/simulation testing.

Bloom's Taxonomy Mapping

Bloom's Level	Activity
Remember	Recall the STEP, DIR, and ENABLE pin names of the A4988 driver and their basic functions.
Understand	Explain why a dedicated driver IC, rather than a direct pin connection, is needed to drive a bipolar stepper motor.

Apply	Compute the commanded angular displacement of the motor shaft from a given step count and step angle.
Analyze	Predict the effect of the DIR logic level and of a mid-sequence DIR change on the direction and consistency of rotation.

Simpson's Taxonomy Mapping (Psychomotor)

Simpson's Level	Activity
Perception	Identify the A4988 driver and stepper motor parts and pins on-screen in Wokwi.
Guided Response	Wire STEP, DIR, and ENABLE (and RESET/SLEEP) to the correct Arduino pins per the connection table.
Mechanism	Run the demonstration sketch and verify motor rotation and direction reversal.
Adaptation	Modify the step count in the sketch to achieve a target commanded angle.

5. Instructor Preparation

Before opening Wokwi (or powering the bench setup), the instructor should be ready to explain the following concepts, connecting each one directly to the STEP/DIR/A4988 circuit used in this experiment.

The Stepper Motor

A stepper motor is an electromechanical actuator that converts a sequence of electrical pulses into discrete, fixed-angle rotations of its shaft, rather than continuous rotation as in a DC motor. A bipolar stepper motor has two independent coils (phases), each with two terminals (1A/1B and 2A/2B), and current direction in each coil must be reversed to advance the rotor — this is why a dedicated driver IC, not a simple transistor switch, is required. A common hybrid stepper has 200 full steps per revolution, giving a step angle of 360°/200 = 1.8° per full step.

The A4988 Driver IC

The Allegro A4988 is a DMOS microstepping driver with a built-in translator: it converts a simple digital STEP/DIR interface into the correct four-phase current sequence for a bipolar stepper, so the microcontroller never has to energise the coils directly. Key pins are summarised below.

Pin	Function
STEP	Rising edge advances the motor by one step (or microstep). Pulse frequency sets rotational speed.
DIR	Logic level sets rotation sense: HIGH is conventionally clockwise, LOW counter-clockwise.
ENABLE	Active-LOW. Driving it LOW turns the output FETs on (motor active); HIGH disables the outputs and de-energises the coils.
MS1, MS2, MS3	Microstep resolution select (full, 1/2, 1/4, 1/8, 1/16 step). Internal pull-downs default to full-step mode if left unconnected.
RESET / SLEEP	Both active-LOW; must be held HIGH for normal operation. RESET is commonly tied to SLEEP so a single HIGH level enables both.
VDD / GND	Logic supply (3–5.5 V) and logic ground, separate from the motor power rail.

1A, 1B, 2A, 2B	Outputs to the two stepper motor coils.

STEP/DIR Timing

Each STEP pulse must be held HIGH and LOW for a minimum pulse width (typically a few microseconds on real hardware) before the next transition; the demonstration sketch in Section 9 uses an 800 µs HIGH/LOW pulse for a clearly observable, safe simulation speed. The DIR pin must be set before the STEP pulse train for a given direction begins, since the driver's translator reads DIR at the moment of each STEP edge.

Steps-to-Angle Relationship

The commanded angular displacement is obtained from the number of STEP pulses and the effective step angle:

$$\theta commanded = nsteps \times \theta step$$

where nsteps is the number of STEP pulses issued and θstep is 1.8° in full-step mode (or 1.8°/m under m-times microstepping, e.g. 0.1125° at 1/16 step).

Worked example: For nsteps = 100 in full-step mode, θ = 100 × 1.8° = 180° (half a revolution).

Open-Loop Position Control

Because a stepper motor is normally driven open-loop (no position feedback), the controller's step count is assumed to equal the actual shaft rotation. This assumption is valid only if the motor never loses synchronism — i.e. it is not stalled by excessive load, speed, or acceleration. Comparing the expected angle (Section 5.4) with the observed angle in Section 11 lets students verify this assumption experimentally.

Objectives

- Understand STEP/DIR-based position and direction control of a bipolar stepper motor.
- Interface the A4988 stepper driver with the Arduino UNO and a bipolar stepper motor.
- Generate STEP pulses and toggle the DIR pin to control rotation direction.
- Relate a commanded step count to the resulting angular displacement of the motor shaft.
- Implement and verify open-loop position control using the A4988 driver.
- Develop confidence in basic driver–actuator integration and STEP/DIR interfacing.

Components Required (Tinkercad/Wokwi and Hardware)

No.	Component	Quantity	Purpose
1	**Arduino UNO**	1	Microcontroller board; generates the STEP and DIR signals and controls the A4988 driver
2	**A4988 Stepper Motor Driver Module**	1	Converts the STEP/DIR digital interface into the correct four-phase current sequence for the motor coils
3	**Bipolar Stepper Motor (e.g., NEMA-17, 200 steps/rev)**	1	Actuator under test; converts commanded STEP pulses into discrete angular rotation

4	**Breadboard**	1	Circuit assembly
5	**Jumper Wires**	As required	Interconnections between Arduino, A4988 driver, and stepper motor
6	**USB Cable / Wokwi account**	1	Programming, power, and Serial Monitor access

Arduino Program

```
const int STEP = 3, DIR = 4, EN = 5;

void moveSteps(long n, bool d) {
  digitalWrite(DIR, d);
  for (long i = 0; i < n; i++) {
    digitalWrite(STEP, HIGH);
    delayMicroseconds(800);
    digitalWrite(STEP, LOW);
    delayMicroseconds(800);
  }
}

void setup() {
  pinMode(STEP, OUTPUT);
  pinMode(DIR, OUTPUT);
  pinMode(EN, OUTPUT);
  digitalWrite(EN, LOW);   // LOW enables the A4988 outputs
}

void loop() {
  moveSteps(100, true);    // 100 steps clockwise (~180 deg, full-step)
  delay(1000);
  moveSteps(100, false);   // 100 steps counter-clockwise
  delay(1000);
}
```

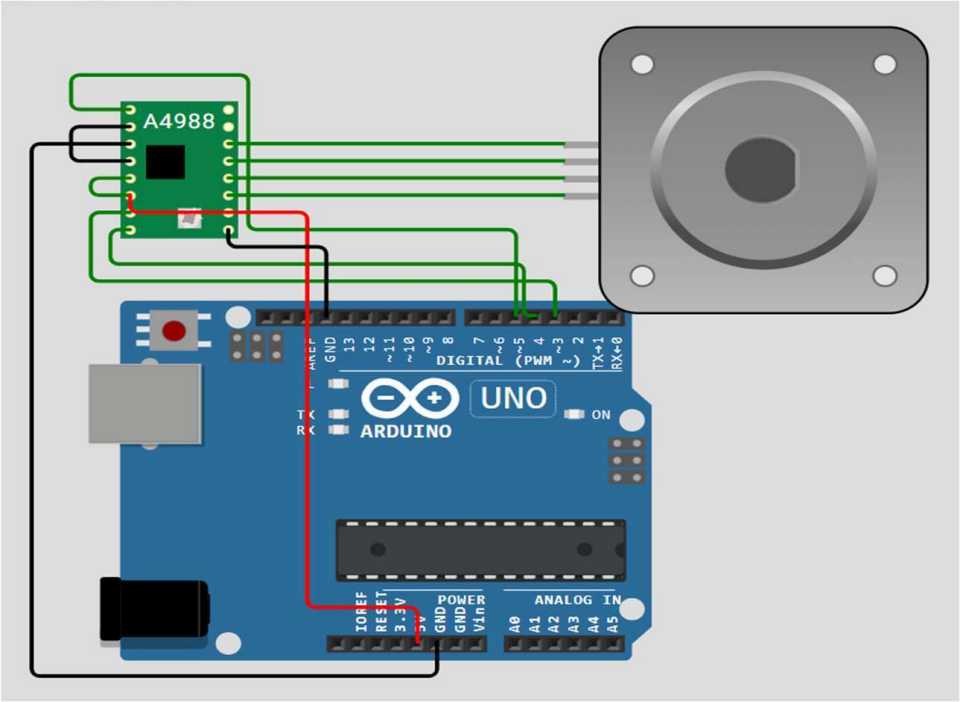


Figure 1. A4988 STEP/DIR interface between Arduino UNO and a bipolar stepper motor.

Connection Table

Signal / Component	Arduino Pin	Connection Details
STEP	D3	A4988 STEP input
DIR	D4	A4988 DIR input
ENABLE	D5	A4988 ENABLE input
A4988 VDD	5V	Logic supply
A4988 GND	GND	Common ground
Motor outputs	—	1A, 1B, 2A, 2B → two bipolar motor coils
RESET	—	Tied to SLEEP (pulled HIGH)
SLEEP	5V	Held HIGH to keep driver enabled
M1, M2, M3 (shorted)	GND	Full-step mode selected

6. Software & Resources Required

Resource	Purpose
Wokwi (wokwi.com)	Primary browser-based simulator for the A4988 + stepper motor circuit.
Arduino IDE / Wokwi code editor	Writing, compiling and uploading/simulating the STEP/DIR control sketch.

Arduino UNO, A4988 module, bipolar stepper motor, breadboard, jumper wires	Physical bench implementation (Section 9), optional/parallel to simulation.
Observation Sheet	Student record of direction, step count, and expected/observed angle.

This experiment follows a Hardware-Integrated and Simulation-Based approach. Physical components, driver modules, and Arduino-based hardware are used for hands-on implementation and verification, while Wokwi is used for circuit design, programming, simulation, and STEP/DIR Position and Direction Control testing.

7. Student Preparation (Pre-Lab)

- Review Experiment 3 notes on DC motor / actuator interfacing with Arduino, and create a free Wokwi account prior to the session or join the instructor's shared Wokwi project using the provided class code.
- Review the pre-lab material on bipolar stepper motors, the A4988 driver, the STEP/DIR/ENABLE interface, and the steps-to-angle relationship before the session.
- Bring a notebook to record circuit connections, commanded step counts, direction settings, expected and observed angles, and experimental observations.

8. Minute-wise Instructor & Student Activity Plan

Time	Instructor Activity	Student Activity	Domain
0–20 min	Introduce the experiment, aim, objectives, laboratory/simulation practices, safety requirements, and applications of stepper motor position and direction control in robotics.	Listen to the introduction, understand the importance of safe laboratory/simulation practices, and show readiness to participate in the activity.	Affective (Heart)
20–60 min	Explain the working principle of the bipolar stepper motor, construction of its two coils, and why a dedicated driver IC (rather than a direct pin connection) is required. Discuss practical applications of STEP/DIR position control.	Understand the construction and operating principle of a bipolar stepper motor and explain its role in 3D printers, CNC machines, robotic arms, and camera gimbals.	Cognitive (Head)
60–95 min	Explain the A4988 driver, its STEP, DIR, ENABLE, MS1/MS2/MS3, RESET, and SLEEP pins, and the relationship between STEP pulses, DIR level, and shaft rotation.	Identify the A4988 driver pins and the stepper motor terminals, and explain the function of each pin/terminal in the position-control system.	Cognitive (Head)
95–135 min	Demonstrate the Arduino UNO, A4988 driver, bipolar stepper motor, and breadboard identification. Show STEP, DIR, ENABLE, VDD, GND, RESET/SLEEP, and motor-coil	Identify the components and terminals, arrange the components, build the circuit, and verify the required power, signal, and common-ground connections.	Psychomotor (Hand)

	connections in Wokwi or hardware.		
135–180 min	Explain the Arduino program using digitalWrite(), delayMicroseconds(), the moveSteps() function, the DIR/STEP pulse sequence, and the steps-to-angle relationship.	Interpret the program statements and explain the complete sequence: DIR set → STEP pulses generated → motor advances → commanded angle achieved.	Cognitive (Head)
180–220 min	Demonstrate program entry/upload and guide students in running the STEP/DIR control program in simulation or hardware.	Enter/upload the Arduino program, start the simulation or hardware test, and verify clockwise and counter-clockwise rotation of the motor shaft.	Psychomotor (Hand)
220–260 min	Guide systematic experimentation using different commanded step counts and demonstrate how to compare the expected angle with the observed shaft rotation.	Command different step counts and directions, record at least five sets of direction, step count, expected angle, and observed angle.	Psychomotor (Hand)
260–302 min	Explain the steps-to-angle calculation ($\theta = nsteps \times \theta step$), STEP/DIR pulse timing constraints, open-loop control assumptions, sensor/motor limitations, and troubleshooting logic.	Analyze the effect of step count and DIR level on system response, compare expected and observed angles, and identify possible causes of errors involving wiring, driver enable state, code, power, or step loss.	Cognitive (Head)
302–326 min	Guide students in modifying the step count and direction to command different target angles and testing the resulting motor response in simulation or hardware.	Modify the program to command a target angle, test different step counts and directions, verify motor response, and correct simple wiring or programming errors.	Psychomotor (Hand)
326–360 min	Discuss safe operation, responsible handling of electronic components, teamwork, and practical applications of stepper motor position control. Conduct review/Q&A and provide post-lab instructions.	Participate in discussion and Q&A, demonstrate safety awareness and teamwork, summarize the experiment, and note the post-lab work.	Affective (Heart)

9. Instructor Demonstration Steps (Simulation and Hardware Implementation)

Step	Procedure
Step 1	Open Wokwi and create a new Arduino UNO project.
Step 2	Add an A4988 driver part and a Stepper Motor part; short M1, M2, and M3 to GND for full-step mode.
Step 3	Wire STEP → D3, DIR → D4, ENABLE → D5, VDD → 5V, GND → GND, RESET–SLEEP tied together to 5V, and motor coils to 1A/1B/2A/2B, exactly as in Figure 1.

Step 4	Paste the demonstration sketch (Section 10.1) into the code editor.
Step 5	Click Start Simulation; observe the shaft rotate 100 steps clockwise, pause, then 100 steps counter-clockwise.
Step 6	Change the step count and re-run; ask students to predict the new angle before observing it.

10. Guided / Probing Questions

- Why can a stepper motor coil not be driven directly from an Arduino digital pin?
- What happens to the motor if the ENABLE pin is held HIGH instead of LOW?
- If DIR is toggled while STEP pulses are being sent, what problem could occur?
- Why might the observed angle differ slightly from the expected angle in a physical setup, even though it matches exactly in simulation?
- The A4988 is wired with STEP on D3 and DIR on D4, but the program uses different pin numbers. What program modification is required, and what happens if it is not corrected?
- What could cause the motor to fail to rotate even though the sketch uploads and runs without error?
- Why must RESET and SLEEP be held HIGH for normal operation of the A4988?
- How would you modify the program to command a 90° movement in full-step mode?
- Why is open-loop stepper control described as an "assumption" of position?
- What is the purpose of the MS1, MS2, and MS3 pins, and what happens if they are left unconnected?
- Suggest one practical robotic or automation application in which STEP/DIR position control can be used.

11. Hands-on Activity (Student Task)

Task: Interface the A4988 driver with a bipolar stepper motor and Arduino UNO, and verify STEP/DIR position control using Wokwi and Hardware.

- Log in and create a project named Stepper_A4988_[YourName]_[RollNo].
- Place an Arduino UNO, A4988 driver, bipolar stepper motor, breadboard, and jumper wires, and complete the required circuit connections as per Figure 1 and the connection table.
- Upload the demonstration sketch; confirm the motor rotates in both directions.
- Modify the sketch to command a 90° movement and verify the step count used.
- Implement the hardware circuit with proper precautions while handling the Arduino, A4988 driver, stepper motor, and electrical connections.
- Repeat for 3 different step counts; record direction, steps, expected angle, and observed angle in the Observation Sheet (Section 16).
- Save the project and keep the Wokwi project link ready for instructor review.
- Take a snapshot/video of the working hardware demonstrating direction and position control for instructor review.

12. Discussion Points

- **Real-World Applications of Stepper Motors –** Discuss the use of stepper motors and STEP/DIR position control in 3D printers, CNC machines, camera gimbals, and robotic arms.

- **Full-Step vs Microstepping Trade-off –** Discuss the trade-off between full-step operation (fast, coarse) and microstepping (smooth, but requires a higher pulse rate).
- **Good Lab Practice and Safe Operation –** Discuss verifying pin assignments before power-up, common grounding, and safe current limits for physical hardware, along with the use of Wokwi for simulation-first verification.

13. Reflection Questions (End of Session)

- What is one new concept about stepper motors you learned today?
- Which part of the A4988 wiring did you find confusing, and how did you resolve it?
- How confident do you feel relating a step count to a rotation angle (rate 1–5), and why?

14. Home Assignment

- Command exactly a 90° movement in full-step mode; state and justify the step count used.
- Add a potentiometer on an analog pin to vary the STEP delay (motor speed) in code.
- Read the short section on microstepping (MS1/MS2/MS3 truth table) in preparation for the next experiment.

15. Assessment Rubrics

Rating Scale: 4 = Excellent, 3 = Good, 2 = Satisfactory, 1 = Needs Improvement (aligned to session domain weightage: Cognitive 30% / Psychomotor 55% / Affective 15%)

Criterion	Domain	4 – Excellent	3 – Good	2 – Satisfactory	1 – Needs Improvement
Concept of STEP/DIR Interfacing	Cognitive	Explains STEP, DIR, ENABLE correctly with justification	Explains most pin functions correctly	Explains with support	Unable to explain
Circuit Wiring	Psychomotor	Independently wires driver + motor correctly	Minor errors, minimal help	Wired with considerable help	Incomplete / incorrect wiring
Program & Verification	Psychomotor	Sketch runs; direction reversal and angle verified accurately	Runs with minor errors	Runs with instructor help	Non-functional program
Lab Engagement	Affective	Actively participates, supports peers	Participates regularly	Participates when prompted	Disengaged

16. Student Observation Sheet (Template)

Sl. No.	Direction	Steps	Expected Angle	Observed Angle
1				

2				
3				
4				
5				

17. Expected Learning Outcome

At the end of this session, a student should be able to independently wire an A4988 driver to a bipolar stepper motor and an Arduino UNO, write a STEP/DIR control sketch, run it in Wokwi (and/or on hardware), correctly relate a commanded step count to the resulting angular displacement of the motor shaft, and explain in their own words how STEP/DIR-based position and direction control is used in robotic and automation applications.

18. Instructor Tips

- Keep the first 35 minutes conceptual; the majority of the session (55%) must remain hands-on wiring and coding.
- Emphasise that ENABLE is active-LOW — this is the most common student wiring error.
- Warn that RESET and SLEEP are both active-LOW and must be tied HIGH, or the driver will ignore all STEP pulses.
- If moving to physical hardware, stress current-limiting (VREF/potentiometer) and common ground before applying motor power.
- Encourage students to predict the expected angle before running the simulation, reinforcing Section 5.4.

CSE 4571 — Practical Robotics Project with Arduino

Basic Questions

No.	Question	Learning Domain	Taxonomy Level
1	What is a stepper motor?	Cognitive	Bloom's Taxonomy – Remember
2	What do STEP and DIR mean in the A4988 interface?	Cognitive	Bloom's Taxonomy – Remember
3	Why is a dedicated driver IC used instead of connecting the motor coils directly to Arduino pins?	Cognitive	Bloom's Taxonomy – Understand
4	What is the function of the ENABLE pin, and what logic level activates the driver?	Cognitive	Bloom's Taxonomy – Understand
5	What is the step angle of a standard 200-step-per-revolution motor?	Cognitive	Bloom's Taxonomy – Remember
6	Calculate the commanded angle for 100 full steps.	Cognitive	Bloom's Taxonomy – Apply
7	What is the purpose of the MS1, MS2, MS3 pins?	Cognitive	Bloom's Taxonomy – Understand
8	Why must RESET and SLEEP be held HIGH for normal operation?	Cognitive	Bloom's Taxonomy – Understand
9	What could cause the observed angle to differ from the expected angle on physical hardware?	Cognitive	Bloom's Taxonomy – Analyze
10	Why is open-loop stepper control described as an "assumption" of position?	Cognitive	Bloom's Taxonomy – Analyze
11	Can you locate and add the A4988 and stepper motor parts in Wokwi?	Psychomotor	Simpson's Taxonomy – Perception
12	Can you identify the STEP, DIR and ENABLE pins on the A4988 module?	Psychomotor	Simpson's Taxonomy – Perception
13	Can you wire STEP, DIR and ENABLE to the correct Arduino pins?	Psychomotor	Simpson's Taxonomy – Guided Response
14	Can you connect RESET and SLEEP correctly to keep the driver enabled?	Psychomotor	Simpson's Taxonomy – Guided Response
15	Can you run the simulation and verify direction reversal?	Psychomotor	Simpson's Taxonomy – Mechanism
16	Can you modify the step count to achieve a target angle?	Psychomotor	Simpson's Taxonomy – Adaptation
17	Are you willing to follow proper wiring-verification practice before running the circuit?	Affective	Krathwohl's Taxonomy – Receiving
18	How actively did you participate and support your peers during the activity?	Affective	Krathwohl's Taxonomy – Responding

19	Do you value simulation-first verification before physical hardware testing? Why?	Affective	Krathwohl's Taxonomy – Valuing

ITER | Siksha 'O' Anusandhan (Deemed to be University)